



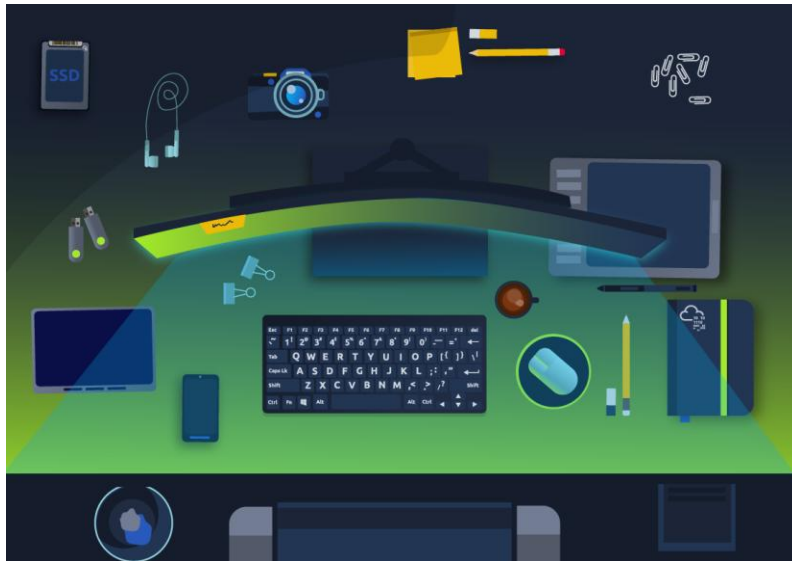
Intro to Digital Forensics

Task 1 – Introduction To Digital Forensics

Forensics is the application of science to investigate crimes and establish facts. With the use and spread of digital systems, such as computers and smartphones, a new branch of forensics was born to investigate related crimes: computer forensics, which later evolved into *digital forensics*.

Think about the following scenario. The law enforcement agents arrive at a crime scene; however, part of this crime scene includes digital devices and media. Digital devices include desktop computers, laptops, digital cameras, music players, and smartphones, to name a few. Digital media includes CDs, DVDs, USB flash memory drives, and external storage. A few questions arise:

- How should the police collect digital evidence, such as smartphones and laptops? What are the procedures to follow if the computer and smartphone are running?
- How to transfer the digital evidence? Are there certain best practices to follow when moving computers, for instance?
- How to analyze the collected digital evidence? Personal device storage ranges between tens of gigabytes to several terabytes; how can this be analyzed?



Assuming this employee is suspected in the figure above, we can quickly see the digital devices that might be of interest to an investigation. We notice a tablet, a smartphone, a digital camera, and a USB flash memory in addition to a desktop computer. Any of these devices might contain a trove of information that can help with an investigation. Processing these as evidence would require digital forensics.

More formally, digital forensics is the application of computer science to investigate digital evidence for a legal purpose. Digital forensics is used in two types of investigations:

1. **Public-sector investigations** refer to the investigations carried out by government and law enforcement agencies. They would be part of a crime or civil investigation.
2. **Private-sector investigations** refer to the investigations carried out by corporate bodies by assigning a private investigator, whether in-house or outsourced. They are triggered by corporate policy violations.

Whether investigating a crime or a corporate policy violation, part of the evidence is related to digital devices and digital media. This is where digital forensics comes into play and tries to establish what has happened. Without trained digital forensics investigators, it won't be possible to process any digital evidence properly.



Answer the questions below

Consider the desk in the photo above. In addition to the smartphone, camera, and SD cards, what would be interesting for digital forensics?

The laptop would also be highly relevant in a digital forensics investigation. It may contain user documents, browser history, emails, saved credentials, deleted files, system logs, and evidence of activity connected to the other devices.

Answer: Laptop

[illegible]

As a digital forensics investigator, you arrive at a scene similar to the one shown in the image above. What should you do as a digital forensics investigator? After getting the proper legal authorization, the basic plan goes as follows:

1. Acquire the evidence: Collect the digital devices such as laptops, storage devices, and digital cameras. (Note that laptops and computers require special handling if they are turned on; however, this is outside the scope of this room.)
2. Establish a chain of custody: Fill out the related form appropriately ([Sample form](#)). The purpose is to ensure that only the authorized investigators had access to the evidence and no one could have tampered with it.
3. Place the evidence in a secure container : You want to ensure that the evidence does not get damaged. In the case of smartphones, you want to ensure that they cannot access the network, so they don't get wiped remotely.
4. Transport the evidence to your digital forensics lab.

At the lab, the process goes as follows:

1. Retrieve the digital evidence from the secure container.
2. Create a forensic copy of the evidence: The forensic copy requires advanced software to avoid modifying the original data.
3. Return the digital evidence to the secure container: You will be working on the copy. If you damage the copy, you can always create a new one.
4. Start processing the copy on your forensics workstation.

The above steps have been adapted from [Guide to Computer Forensics and Investigations, 6th Edition](#).

More generally, according to the former director of the Defense Computer Forensics Laboratory, Ken Zatyko, digital forensics includes:

- Proper search authority: Investigators cannot commence without the proper legal authority.
- Chain of custody: This is necessary to keep track of who was holding the evidence at any time.
- Validation with mathematics: Using a special kind of mathematical function, called a hash function, we can confirm that a file has not been modified.
- Use of validated tools: The tools used in digital forensics should be validated to ensure that they work correctly. For example, if you are creating an image of a disk, you want to ensure that the forensic image is identical to the data on the disk.
- Repeatability: The findings of digital forensics can be reproduced as long as the proper skills and tools are available.
- Reporting: The digital forensics investigation is concluded with a report that shows the evidence related to the case that was discovered.

Answer the questions below

It is essential to keep track of who is handling it at any point in time to ensure that evidence is admissible in the court of law. What is the name of the documentation that would help establish that?

*The documentation is called the **chain of custody**. It records who collected, handled, transferred, examined, and stored the evidence, helping show that it was not altered or tampered with.*

Answer: Chain of custody

Task 3 – Practical Example of Digital Forensics

Everything we do on our digital devices, from smartphones to computers, leaves traces. Let's see how we can use this in the subsequent investigation.

Our cat, Gado, has been kidnapped. The kidnapper has sent us a document with their requests in MS Word Document format. We have converted the document to PDF format and extracted the image from the MS Word file for your convenience.

Document Metadata

When you create a text file, TXT, some metadata gets saved by the Operating System, such as file creation date and last modification date. However, much information gets kept within the file's metadata when you use a more advanced editor, such as MS Word. There are various ways to read the file metadata; you might open them within their official viewer/editor or use a suitable forensic tool. Note that exporting the file to other formats, such as PDF, would maintain most of the metadata of the original document, depending on the PDF writer used.

Let's see what we can learn from the PDF file. We can try to read the metadata using the program pdftinfo. Pdftinfo displays various metadata related to a PDF file, such as title, subject, author, creator, and creation date. (The AttackBox already has pdftinfo installed; however, if you are using Kali Linux and don't have pdftinfo installed, you can install it using

sudo apt install poppler-utils.) Consider the following example of using pdftinfo DOCUMENT.pdf.

Answer the questions below

Using pdftinfo, find out the author of the attached PDF file, ransom-letter.pdf.

```
root@ip: ~# cd /root/Rooms/introdigitalforensics/
root@ip: ~/Rooms/introdigitalforensics# ls
letter-image.jpg  ransom-letter.doc  ransom-letter.pdf  ransom-lettter-2.zip
root@ip: ~/Rooms/introdigitalforensics# pdftinfo ransom-letter.pdf
Title:          Pay NOW
Subject:        We Have Gato
Author:         Ann Gree Shepherd
Creator:        Microsoft® Word 2016
Producer:       Microsoft® Word 2016
CreationDate:   Wed Feb 23 09:10:36 2022 UTC
ModDate:        Wed Feb 23 09:10:36 2022 UTC
Custom Metadata: no
Metadata Stream: yes
Tagged:         yes
UserProperties: no
Suspects:       no
Form:           none
JavaScript:     no
Pages:          1
Encrypted:      no
Page size:      595.44 x 842.04 pts (A4)
Page rot:       0
File size:      71371 bytes
Optimized:      no
PDF version:    1.7
```

The author is:

Answer: Ann Gree Shepherd

Photo EXIF Data

EXIF stands for Exchangeable Image File Format; it is a standard for saving metadata to image files. Whenever you take a photo with your smartphone or with your digital camera, plenty of information gets embedded in the image. The following are examples of metadata that can be found in the original digital images:

- Camera model / Smartphone model
- Date and time of image capture
- Photo settings such as focal length, aperture, shutter speed, and ISO settings

Because smartphones are equipped with a GPS sensor, finding GPS coordinates embedded in the image is highly probable. The GPS coordinates, i.e., latitude and longitude, would generally show the place where the photo was taken.

There are many online and offline tools to read the EXIF data from images. One command-line tool is exiftool. ExifTool is used to read and write metadata in various file types, such as JPEG images. (The AttackBox already has exiftool installed; however, if you are using Kali Linux and don't have exiftool installed, you can install it using `sudo apt install libimage-exiftool-perl`.) In the following terminal window, we executed `exiftool IMAGE.jpg` to read all the EXIF data embedded in this image.

If you take the above coordinates and search one of the online maps, you will learn more about this location. Searching [Microsoft Bing Maps](#) or [Google Maps](#) for 51 deg 30' 51.90" N, 0 deg 5' 38.73" W reveals the street where the photo was taken. Note that for the search to work, we had to replace deg with ° and remove the extra white space. In other words, we typed 51°30'51.9"N 0°05'38.7"W in the map search bar.

Using exiftool or any similar tool, try to find where the kidnappers took the image they attached to their document. What is the name of the street?

```
letter-image.jpg  ransom-letter.doc  ransom-letter.pdf  ransom-lettter-2.zip
root@ip-__ --- --- ---:~/Rooms/introdigitalforensics# exiftool letter-image.jpg
ExifTool Version Number      : 12.76
File Name                    : letter-image.jpg
Directory                    : .
File Size                    : 127 kB
File Modification Date/Time   : 2022:02:23 08:53:33+00:00
File Access Date/Time        : 2026:04:28 14:06:39+00:00
File Inode Change Date/Time   : 2026:02:25 17:59:37+00:00
File Permissions              : -rwxr-xr-x
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                  : 1.01
```



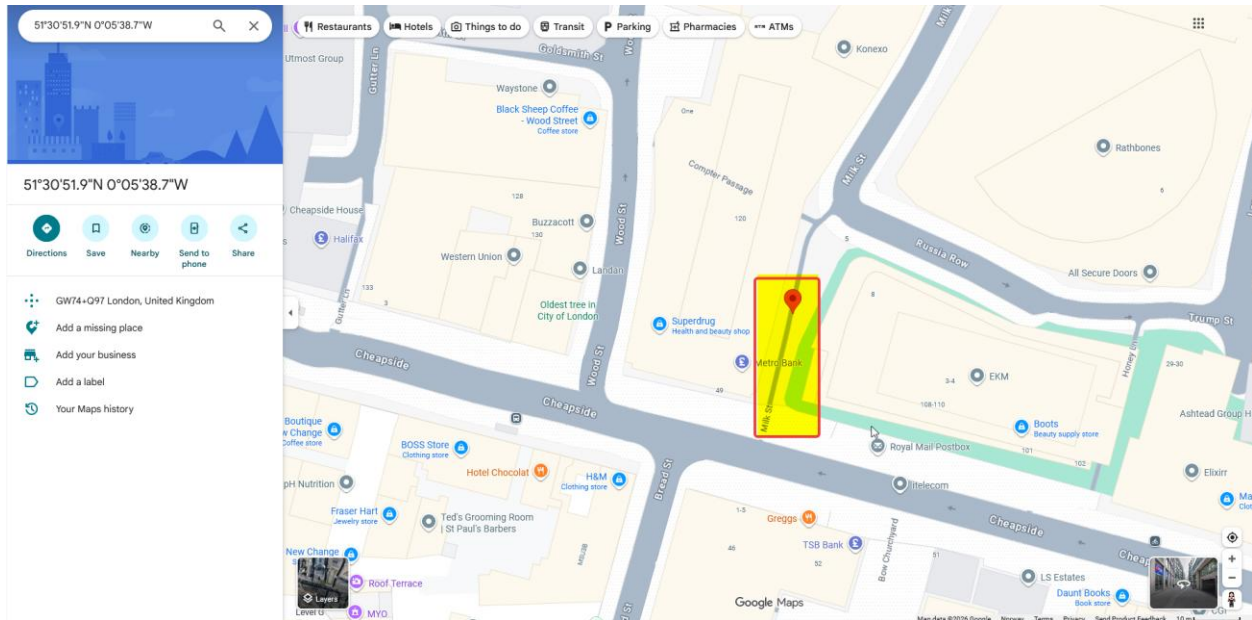
```

GPS Latitude       : 51 deg 30' 51.90" N
GPS Longitude      : 0 deg 5' 38.73" W
Date/Time Created  : 2022:02:15 17:23:40-17:23
Digital Creation Date/Time : 2021:11:05 14:06:13+03:00
Circle of Confusion : 0.043 mm
Depth of Field     : 0.06 m (0.76 - 0.82 m)
Field Of View      : 54.9 deg
Focal Length       : 50.0 mm (35 mm equivalent: 34.6 mm)
GPS Position       : 51 deg 30' 51.90" N, 0 deg 5' 38.73" W
Hyperfocal Distance : 20.58 m
Light Value        : 7.9
Lens ID            : Canon EF 50mm f/1.8 STM

```

Google Maps:

https://www.google.com/maps/place/51%C2%B030'51.9%22N+0%C2%B005'38.7%22W/@51.51442,-0.0966666,17z/data=!3m1!4b1!4m4!3m3!8m2!3d51.5144167!4d-0.0940917?entry=tту&g_ep=EgoyMDI2MDYxNi4wIKXMDSOASAFQAw%3D%3D



The EXIF metadata in letter-image.jpg contained GPS coordinates. Entering these coordinates into Google Maps showed that the image was taken on Milk Street in London.

Answer: Milk Street

What is the model name of the camera used to take this photo?

```

Make : Canon
Camera Model Name : Canon EOS R6

```

Answer: Canon EOS R6

Disclaimer

This writeup was created for educational and portfolio purposes only. All testing was performed in the relevant lab environment.

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